Hydro Unit Commitment With a Head-Sensitive Reservoir and Multiple Vibration Zones Using MILP

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***Abstract*—A batch of huge hydropower plants with multiple vibration zones have been developed in China. These vibration zones varying with net head, i.e., head-sensitive, seriously challenge the hydro unit commitment (HUC). This paper presents a MILP model for HUC with a head-sensitive reservoir and multiple vibration zones. The focus of the study is mainly on two net head related constraints, namely the unit performance curves and multiple vibration zones. These constraints are linearized through the use of binary 0-1 integer variables. The practical behavior of the MILP model is demonstrated using real-world case studies. The results demonstrate that the proposed model is computationally efficient and suitable for large hydropower plants with multiple vibration zones, showing its potential practicability and validity for solving the HUC problem.**

***Index Terms*—Hydro unit commitment, hydropower, head effect, mixed-integer linear programming, multiple vibration zones.**

NOMENCLATURE

The notations used in the model are as follows:

Sets:

I	Set of indices of units.
$\|K_i\|$	Set of indices of safe operating zones of unit i.
N	Set of indices of net head intervals.
T	Set of indices of periods.

Constants:

Δt	Time period [h].
$a_0, \ldots, a_4$	Coefficients dependent on the relationship of reservoir fore-bay water level and water volume.
$b_0, \ldots, b_4$	Coefficients dependent on the relationship of reservoir tailrace water level and water release.
c_i	Coefficient that dependent on the unit penstock characteristics [s^2/m^5].
D_t	Load demand in period t [MW].
$DA^n_{i,k}, DB^n_{i,k}$	Slope and constant term of lower bound of safe operating zone k of unit i for interval n [MW/m], [MW].
H_n	Right endpoint of net head for interval n [m].
IN_t	Inflow in period t [m^3/s].
M_1, M_2	Penalty coefficient.
$\underline{P_i}, \overline{P_i}$	Minimum and maximum power output of unit i [MW].
$\underline{Q_i}, \overline{Q_i}$	Minimum and maximum turbined outflow of unit i [m^3/s].
$\overline{S}$	Maximum spillage from the reservoir [m^3/s].
TA_i	Minimum online time of unit i [h].
TB_i	Minimum offline time of unit i [h].
$\underline{U}, \overline{U}$	Minimum and maximum water release from the reservoir [m^3/s].
$UA^n_{i,k}, UB^n_{i,k}$	Slope and constant term of upper bound of safe operating zone k of unit i for interval n [MW/m], [MW].
V_{beg}	Initial value of water volume in the reservoir [m^3].
$\underline{V}, \overline{V}$	Minimum and maximum water volume in the reservoir [m^3].
$\overline{Y_i}$	Maximum number of startups of unit i over the planning horizon.
δ_i	Approximation error between $p_{i,t}$ and actual power output of unit i [MW].

Variables:

d^n_t	Binary variable, equal to 1 if $h_{i,t}$ is in interval n and 0 otherwise.
$\overline{f^n}_{i,k}, \underline{f^n}_{i,k}$	Upper and lower bound of safe operating zone k of unit i for interval n [MW].
$g_{i,t}$	Binary variable, equal to 1 if unit i is online in period t and 0 otherwise.
$h_{i,t}$	Net head of unit i in period t [m].
$h^{loss}_{i,t}$	Penstock head loss of unit i in period t [m].
$p_{i,t}$	Power output of unit i in period t [MW].
$\overline{p}^o_{i,t}, \underline{p}^o_{i,t}$	Upper and lower bound of vibration zone o of unit i in period t [MW].
$q_{i,t}$	Turbined outflow of unit i in period t [m^3/s].
$r_{i,t,k}$	Binary variable, equal to 0 if power output is in the safe operating zone k of unit i in period t.
s_t	Total reservoir spillage in period t [m^3/s].
u_t	Total water release in period t [m^3/s].
v_t	Water volume in the reservoir at the end of period t [m^3].

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$y_{i,t}$	Binary variable, equal to 1 if unit i is started up and 0 otherwise.
$\widetilde{y}_{i,t}$	Binary variable, equal to 1 if unit i is shut down and 0 otherwise.
z_t	Reservoir fore-bay water level at the end of period t [m].
zd_t	Reservoir tailrace water level in period t [m].

Functions:

$f_{hq}(q_{i,t})$	Penstock head loss as a function of unit turbined outflow [m].
$f_{pqh}(q_{i,t}, h_{i,t})$	Power output as a function of unit turbined outflow and net head.
$f^n_{pq\|h}(q_{i,t})$	Power output as a function of unit turbined outflow for unit performance curve n.
$f_{zu}(u_t)$	Reservoir tailrace water level as a function of total water release.
$f_{zv}(v_t)$	Reservoir fore-bay water level as a function of water volume.

I. Introduction

CHINA is endowed with large hydro potential. During the past two decades, in particular the past decade, there has been a rapid rate of development of hydropower in China. By the end of 2014, China has reached a hydropower generation capacity of 301.83 GW (22% of its estimated total generation capacity), making it the first hydropower generator in the world. Hydropower development in China has included implementation of a batch of huge hydropower projects with large unit capacity of more than 500 MW and high head (above 100 m). Several vibration zones may be surpassed when these units are faced with a complicated task of regulating system loads, imposing a tremendous challenge to the operation and management of hydropower in China [1]. Thus, unit commitment of huge hydropower plants has become increasingly important in the daily operation of power grids. The purpose of hydro unit commitment (HUC) is to optimize the operation of the generating units within a multiunit hydropower plant for a given planning horizon, typically one day or one week. The schedules are determined so as to minimize the total operational cost or total release water while satisfying all reservoir and unit constraints. HUC is an important optimization task in the daily operation planning of modern power system because of its potential benefits.

HUC belonging to unit commitment (UC), is a large-scale, mixed-integer, combinational and nonlinear programming problem [2]. Various methods have been used to solve this type of problem, including priority listing (PL) [3], [4], dynamic programming (DP) [5]–[7], Lagrangian relaxation (LR) [8]–[10], as well as a suit of heuristic algorithms such as simulated annealing (SA) [11], [12], artificial neural networks (ANN) [13], [14], genetic algorithm (GA) [15], [16], tabu search (TS) [17] and evolutionary programming (EP) [18], [19]. The combinations of the aforementioned methods [20]–[25] have been also used to solve the HUC problem. Extensive reviews of the UC literature can be found in [26]–[28].

Mixed-integer linear programming (MILP) has attracted more attention in recent years for solving HUC problems [29]–[35]. This is due to the fact that MILP allows for adding the constraints, incorporating the nonlinearities using piecewise linear approximation, and the introduction of the discrete nature of the problem via adding integer variables or constraints [29]. Nonetheless, modeling nonlinear characteristics through MILP techniques is still challenging. In unit performance curves, power output nonlinearly depends on the unit turbined outflow and net head. Similarly, limits of power output nonlinearly depend on the net head.

The head effect is important for accurate modeling of the HUC problem and has been the subject of previous studies. In [30] and [31], the piecewise linear performance curve was associated with reservoir storage to model head effect; later, [32] enhanced the solution through two-dimensional considerations of water volume and unit turbined outflow obtained from [30]; in [33], head effect was approximated through meshing and triangulation; [34] used nine data points to define the safe operation zone of a unit and introduced a three-dimensional interpolation technique to accurately model head effect; and in [35], a systematic analysis of the effects of linearization of the nonlinearity on solution feasibility was presented.

Multiple vibration zones are the regions that unit power output has to be permanently avoided. For a unit with vibration zones, its operating zone is broken into several sub-regions in practical operations. The isolated sub-regions form multiple decision spaces, making HUC a multidimensional discontinuous optimization problem [36], [37]. Different from regular shaped zones [22], [34], [35], [38] (Fig. 1(a)), irregular shaped vibration zones [39], [40] (Fig. 1(b)) vary with unit net head. The head-sensitive characteristic of irregular shaped vibration zones makes HUC problem more difficult because of the interrelationships of power output in different time periods [39]. Thus, accurate modeling of multiple irregular shaped vibration zones is essential for solving the HUC problem. In our previous work [39], individual units were converted to a united generator through assembled mathematical techniques and dynamic programming, resulting in a plant-based scheduling model. An optimization framework that combined the progressive optimality algorithm with a vibration zone avoidance strategy was presented. This work extends our previous works from plant-based to unit-based scheduling.

In this paper, we focus on developing an accurate MILP model for HUC with a head-sensitive reservoir and multiple vibration zones. The unit performance curves are discretized into a set of piecewise curves based on discretized net head so that head effect can be modeled. The main contribution of the paper is approximating the multiple vibration zones using several simple polygons depending on discretized net head so that the head effect is captured. The commercial optimization software, LINGO, is adopted and branch and bound algorithm is used to solve the MILP model. Three test cases with actual hydropower plant data are presented. The results demonstrate that the proposed model solves the problem with an acceptable computational burden for a huge hydropower plant with head-sensitive vibration zones, showing its potential practicability and validity for solving the HUC problem.

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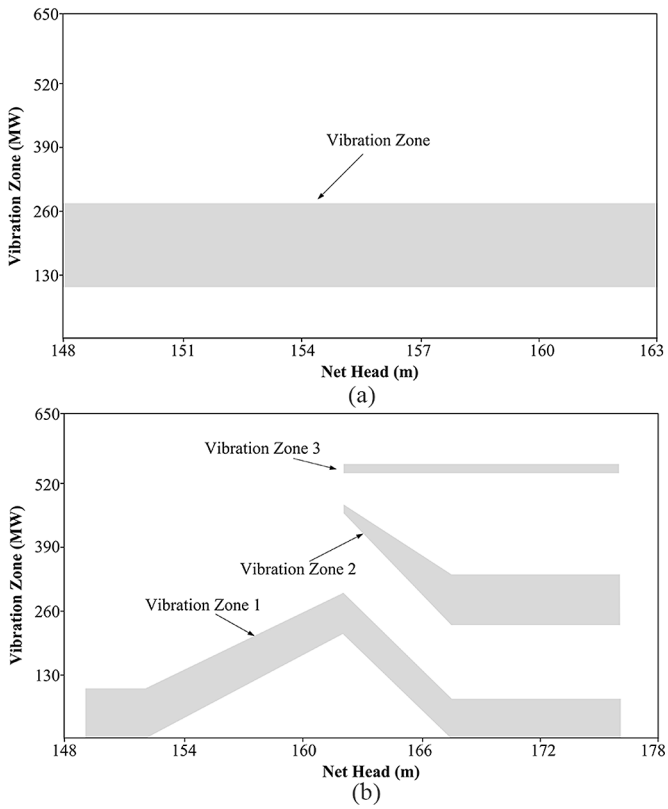


Fig. 1. Vibration illustration. (a) Regular shaped vibration zone, (b) Multiple irregular shaped vibration zones.

The paper is organized as follows. Section II describes the mathematical formulation of the HUC problem. In particular, two main constraints, the unit performance curves and multiple vibration zones, are linearized through MILP techniques. In Section III, results from real-word case studies are provided and discussed. Finally, conclusions are drawn in Section IV.

II. Mathematical Model

A. Problem Formulation

1) Objective Function: The goal of this model is to minimize total water release while meeting the specified load demands over the operation periods. The objective function is represented as

$$\min \sum_{t\in T} u_t \times \Delta t \times 3600 \tag{1}$$

2) Constraints: The constraints of the model can be classified into linear constraints (2)–(18) and nonlinear constraints (19)–(23).

$$u_t = \sum_{i\in I} q_{i,t} + s_t \quad \forall t \tag{2}$$

$$v_t = v_{t-1} + (IN_t - u_t) \times \Delta t \times 3600 \quad \forall t \tag{3}$$

$$\underline{V} \le v_t \le \overline{V} \quad \forall t \tag{4}$$

$$\underline{U} \le u_t \le \overline{U} \quad \forall t \tag{5}$$

$$0 \le s_t \le \overline{S} \quad \forall t \tag{6}$$

$$\sum_{i\in I} p_{i,t} = D_t \quad \forall t \tag{7}$$

$$g_{i,t} \times \underline{P_i} \le p_{i,t} \le g_{i,t} \times \overline{P_i} \quad \forall i, \forall t \tag{8}$$

$$g_{i,t} \in \{0,1\} \quad \forall i, \forall t \tag{9}$$

$$g_{i,t} \times \underline{Q_i} \le q_{i,t} \le g_{i,t} \times \overline{Q_i} \quad \forall i, \forall t \tag{10}$$

$$v_0 = V_{beg} \tag{11}$$

$$y_{i,t} - \widetilde{y}_{i,t} = g_{i,t} - g_{i,t-1} \quad \forall i, \forall t \tag{12}$$

$$y_{i,t} + \widetilde{y}_{i,t} \le 1 \quad \forall i, \forall t \tag{13}$$

$$y_{i,t}, \widetilde{y}_{i,t} \in \{0,1\} \quad \forall i, \forall t \tag{14}$$

$$y_{i,t} + \sum_{\eta=t+1}^{t+TA_i-1} \widetilde{y}_{i,\eta} \le 1 \quad \forall i, \forall t \tag{15}$$

$$\widetilde{y}_{i,t} + \sum_{\eta=t+1}^{t+TB_i-1} y_{i,\eta} \le 1 \quad \forall i, \forall t \tag{16}$$

$$\sum_{t\in T} y_{i,t} \le \overline{Y_i} \quad \forall i \tag{17}$$

$$h_{i,t} = (z_{t-1} + z_t)/2 - zd_t - h_{i,t}^{loss} \quad \forall i, \forall t \tag{18}$$

$$z_t = f_{zv}(v_t) \quad \forall t \tag{19}$$

$$zd_t = f_{zu}(u_t) \quad \forall t \tag{20}$$

$$h_{i,t}^{loss} = f_{hq}(q_{i,t}) \quad \forall i, \forall t \tag{21}$$

$$p_{i,t} = f_{pqh}(q_{i,t}, h_{i,t}) \quad \forall i, \forall t \tag{22}$$

$$(p_{i,t} - \overline{p}_{i,t}^{o})(p_{i,t} - \underline{p}_{i,t}^{o}) > 0 \quad \forall i, \forall t, \forall o \tag{23}$$

Constraint (2) defines the total water release. Constraint (3) is the continuity balance equation. Constraints (4), (5) and (6) establish lower and upper bound on storage, total water release and spillage, respectively. Constraint (7) imposes the load demand balance. Constraints (8)–(10) set lower and upper bounds on power output and unit turbined outflow. Constraint (11) sets the initial water volume. Constraints (12)–(14) set the logical status of unit commitment. Constraints (15) and (16) guarantee minimum online/offline time of a unit. Constraint (17) is used to limit unit startups. Constraint (18) gives the expression of net water head. Constraint (19) defines the relationship of reservoir fore-bay water level and water volume. Constraint (20) defines the relationship of reservoir tailrace water level and total water release. Constraint (21) defines the relationship of penstock head loss and unit turbined outflow. Constraint (22) sets the relationship of power output, unit turbined outflow and net head. Constraint (23) establishes multiple vibration zones.

B. Net Head Linearization

In (18), net head is a function of reservoir fore-bay water level, tailrace water level and penstock head loss. Reservoir fore-bay water level is calculated using a nonlinear function of water volume in the reservoir, and tailrace water level depends nonlinearly on total outflow. The fore-bay and tailrace water levels are usually expressed by a fourth-order polynomial in terms of water volume in the reservoir and total outflow, (24) and (25). The penstock head loss is usually modeled as a

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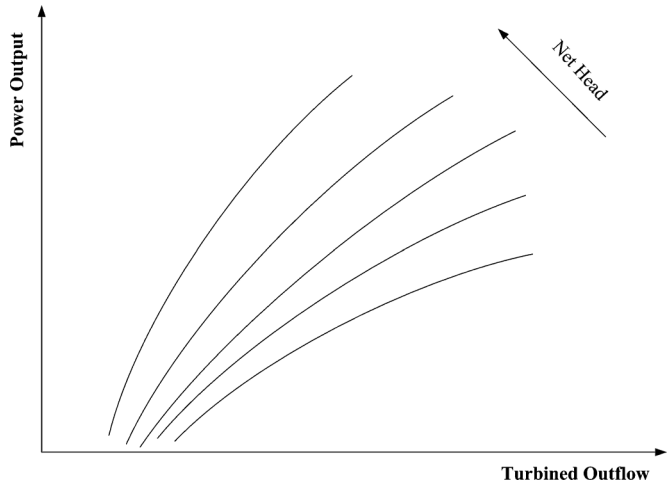


Fig. 2. Unit performance curves.

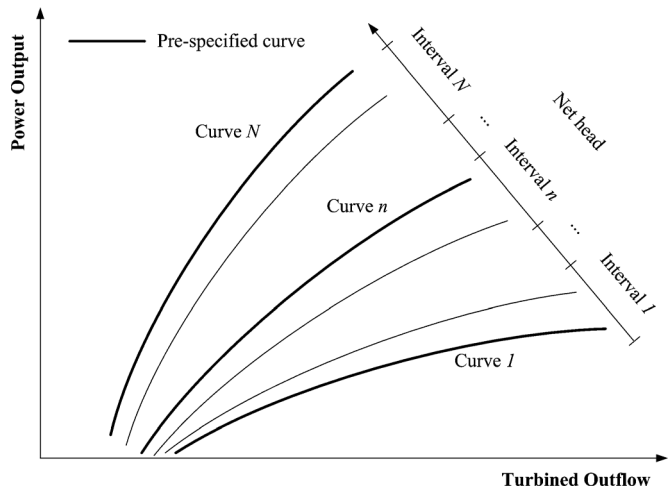


Fig. 3. Simplified unit performance curves.

quadratic function of the unit turbined outflow (26). These nonlinear functions are all approximated by a piecewise linearization technique proposed in [35].

$$z_t = a_0 + a_1 v_t^1 + a_2 v_t^2 + a_3 v_t^3 + a_4 v_t^4 \ \ \forall t \tag{24}$$

$$zd_t = b_0 + b_1 u_t^1 + b_2 u_t^2 + b_3 u_t^3 + b_4 u_t^4 \ \ \forall t \tag{25}$$

$$h_{i,t}^{loss} = c_i q_{i,t}^2 \ \ \forall t, \forall t \tag{26}$$

C. Unit Performance Curves Linearization

In (22), unit performance curves refer to a family of curves representing the nonlinear relationship of power output, unit turbined outflow and net head, as shown in Fig. 2. For a head-sensitive reservoir, net head variation cannot be ignored over the whole scheduling horizon. Taking a reservoir with relatively large capacity as an example, a large change in net head always occurs because of the sensitive relationship between tailrace water level and total water release in spite of the relatively fixed fore-bay water level. Thus, modeling the head effect is crucial for accurate approximation of unit performance curves.

The unit performance curves linearization proposed in this paper contains three steps. In the first step, the allowable net head variation is discretized into a pre-specified number of successive intervals. In the second step, the unit performance curves are simplified into a set of curves, each one for a net head interval, in which case power output is solely dependent on the unit turbined outflow. In the third step, each curve is modeled through a piecewise linear formulation. These steps aim to achieve the accurate modeling of the head effect for approximating unit performance curves.

1) Discretization of the Net Head on a Unit: Here, the allowable net head variation is discretized into a pre-specified number of successive intervals. This discretization is modeled through a classical use of binary variables, which is presented as:

$$\sum_{n\in N} d_t^n \times H_{n-1} \le h_{i,t} \le \sum_{n\in N} d_t^n \times H_n \ \ \forall i, \forall t \tag{27}$$

$$\sum_{n\in N} d_t^n = 1 \ \ \forall t \tag{28}$$

$$d_t^n \in \{0,1\} \ \ \forall t, \forall n \tag{29}$$

In (27)–(29), $(d_t^1, \ldots, d_t^n, \ldots, d_t^N)$ are the binary variables used to choose the right net head interval according to the net water head. According to constraints (28) and (29), all binary variables are equal to 0 but one. If d_t^n is equal to 1, constraint (27) forces $h_{i,t}$ to be in the interval $[H_{n-1}, H_n]$.

2) Simplification of the Unit Performance Curves: As mentioned above, the unit performance curves are three-dimensional. Fig. 3 depicts an illustrative example of the proposed simplification. The unit performance curves are simplified into a pre-specified number of curves, each one for a net head interval. Hence, the three-dimensional unit performance curves are converted to two-dimensional curves, in which case power output solely depends on unit turbined outflow in each net head interval.

The simplification is modeled through the joint use of multiple constraints, expressed as:

$$-\overline{P_i} \times (1 - d_t^1) \le p_{i,t} - f_{pq|h}^1(q_{i,t}) \le \overline{P_i} \times (1 - d_t^1) \quad \forall i, \forall t \tag{30}$$

$$\cdots\cdots$$

$$-\overline{P_i} \times (1 - d_t^n) \le p_{i,t} - f_{pq|h}^n(q_{i,t}) \le \overline{P_i} \times (1 - d_t^n) \quad \forall i, \forall t \tag{31}$$

$$\cdots\cdots$$

$$-\overline{P_i} \times (1 - d_t^N) \le p_{i,t} - f_{pq|h}^N(q_{i,t}) \le \overline{P_i} \times (1 - d_t^N) \quad \forall i, \forall t \tag{32}$$

In the formulation above, constraint (31) represents curve n (see Fig. 3). Due to constraints (28) and (29), all binary variables are equal to 0 but one. If d_t^n is equal to 1, constraints (31) forces power output to be equal to $f_{pq|h}^n(q_{i,t})$ so that curve n is used. In this case, the remaining constraints are deactivated by the inclusion of $\overline{P_i}$. Similarly, constraints (30) and (32), model curves *1* and N, respectively (see Fig. 3).

3) Piecewise Linearization of the Unit Performance Curves: For each water head interval, the two-dimensional unit performance curve is nonlinear. Following the idea in [30], piecewise linearization is used here.

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D. Linearization of Multiple Vibration Zones

In (23), multiple vibration zones refer to the regions that power output has to be permanently avoided. Practical examples are given in Fig. 1. Different from medium- and small-size units with a single regular shaped vibration zone, large-size units have multiple and irregular shaped zones that are closely related to the net head. Thus, modeling of head effect is crucial for accurate approximation of multiple vibration zones. As discussed earlier, taking into account the head effect on multiple vibration zones is this paper's a main contribution.

The linearization of multiple vibration zones proposed in this paper involves two steps. In the first step, the multiple vibration zones are approximated by several simple polygons. In the second step, the avoidance of vibration zones is modeled through a linear formulation. These steps aim to result in an accurate modeling of the head effect for approximating multiple vibration zones.

1) Approximation of Multiple Vibration Zones: Irregular shaped vibration zones imply irregular shaped safe operating zones in which one unit can only operate. Approximating safe operating zones is equivalent to the approximation of vibration zones.

Fig. 4 provides an illustrative example of this approximation. The irregular shaped safe operating zones in each net head interval are approximated by a set of polygonal ones so that the head effect can be considered in each net head interval.

In each net head interval, the upper and lower bounds of safe operating zones are treated as a linear and monotonic function of net head. In practice, this can be achieved by the appropriate discretization of net head. The linearity and monotony are also used to discretize net head in Section II-C-1). The formulations of upper and lower bounds are as following:

$$\overline{f^n}_{i,k} = UA^n_{i,k} \times h_{i,t} + UB^n_{i,k} \; \forall i, \forall n, \forall k \tag{33}$$

$$\underline{f^n}_{i,k} = DA^n_{i,k} \times h_{i,t} + DB^n_{i,k} \; \forall i, \forall n, \forall k \tag{34}$$

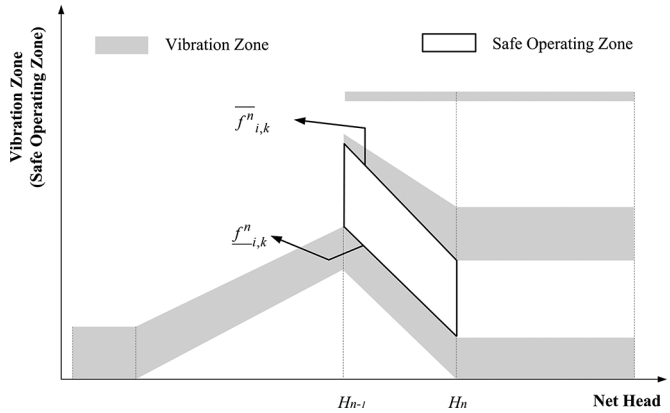


Fig. 4. Approximation of safe operating zones.

2) Linearization of Multiple Vibration Zones Avoidance: The multiple vibration zones constraint (constraint (23)) is treated as a hard constraint in this paper so that power output has to permanently avoided over the entire time horizon. In this way, the power output is limited in safe operating zones over the entire planning horizon. The safe operating zones for each net head interval are modeled using binary variables. The head effect is modeled through a joint use of multiple constraints. Accordingly, constraint (23) can be formulated using the linear constraints in (35)–(42) at the bottom of the page.

The merit of (35)–(42) is in the fact that constraints are independent of the form of the safe operating zone bounds. In (35)–(40), δ_i is an approximation error proposed in [35]. Constraints (37) and (38) represent power output being restricted in the safe operating zones for $[H_{n-1}, H_n]$. Due to constraints (28) and (29), all binary variables are equal to 0 but one. If d^n_t is equal to 1, constraints (37) and (38) are converted to constraint (43), shown at the bottom of the page.

In this case, the remaining constraints are deactivated by the inclusion of M_2. Similarly, constraints (35) and (36), as well as

$$p_{i,t} \geq (\underline{f^1}_{i,k} + \delta_i) - r_{i,t,k} \times M_1 - (1 - d^1_t) \times M_2 \; \forall i, \forall t, \forall k \tag{35}$$

$$p_{i,t} \leq (\overline{f^1}_{i,k} - \delta_i) + r_{i,t,k} \times M_1 + (1 - d^1_t) \times M_2 \; \forall i, \forall t, \forall k \tag{36}$$

$$\cdots\cdots$$

$$p_{i,t} \geq (\underline{f^n}_{i,k} + \delta_i) - r_{i,t,k} \times M_1 - (1 - d^n_t) \times M_2 \; \forall i, \forall t, \forall k \tag{37}$$

$$p_{i,t} \leq (\overline{f^n}_{i,k} - \delta_i) + r_{i,t,k} \times M_1 + (1 - d^n_t) \times M_2 \; \forall i, \forall t, \forall k \tag{38}$$

$$\cdots\cdots$$

$$p_{i,t} \geq (\underline{f^N}_{i,k} + \delta_i) - r_{i,t,k} \times M_1 - (1 - d^N_t) \times M_2 \; \forall i, \forall t, \forall k \tag{39}$$

$$p_{i,t} \leq (\overline{f^N}_{i,k} - \delta_i) + r_{i,t,k} \times M_1 + (1 - d^N_t) \times M_2 \; \forall i, \forall t, \forall k \tag{40}$$

$$\sum_{k \in |K_i|} r_{i,t,k} = |K_i| - g_{i,t} \;\; \forall i, \forall t \tag{41}$$

$$r_{i,t,k} \in \{0, 1\} \; \forall i, \forall t, \forall k \tag{42}$$

$$(\underline{f^n}_{i,k} + \delta_i) - r_{i,t,k} \times M_1 \leq p_{i,t} \leq (\overline{f^n}_{i,k} - \delta_i) + r_{i,t,k} \times M_1 \; \forall i, \forall t, \forall k \tag{43}$$

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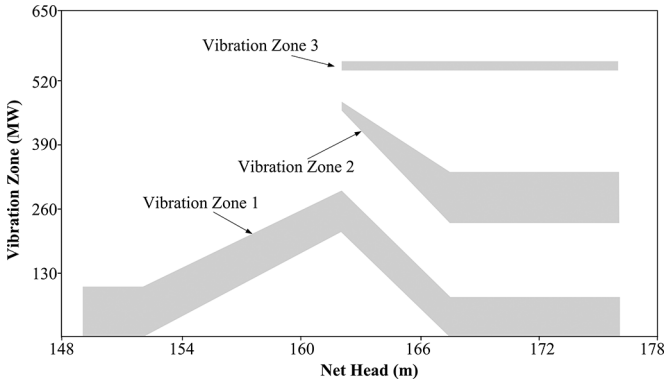


Fig. 5. Multiple vibration zones of unit #2.

TABLE I
MAIN CHARACTERISTICS OF THE GOUPITAN HYDROPOWER PROJECT

Dam height (m)	Storage capacity ($10^8 m^3$)	Installed capacity (MW)	Turbine(MW) (capacity×units)
232.50	64.54	3,000	600×5

(39) and (40), model power output being restricted in the safe operating zone for $[H_0, H_1]$ and $[H_{N-1}, H_N]$, respectively.

Based on (41) and (42), all binary variables are equal to 1 but one. For an online unit, if $r_{i,t,k}$ is equal to 0, the power output is in the safe operating zone $[\underline{f^n}_{i,k}, \overline{f^n}_{i,k}]$. In this case, constraint (43) is converted to constraint (44).

$$(\underline{f^{\mathrm{n}}}_{i,k} + \delta_i) \le p_{i,t} \le (\overline{f^n}_{i,k} - \delta_i) \ \forall i, \forall t \tag{44}$$

III. CASE STUDY

The proposed model in the previous section is solved by LINGO 14.0 [41]. The branch and bound algorithm is called to solve the MILP model. The modeling exercise is performed on a DELL T1600 workstation containing an Inter (R) Xeon (R) CPU with 3.30 GHz and 4.0 GB of RAM.

The proposed model is applied to a real-world case to evaluate its practicality. The Goupitan hydropower project is located on the Wujiang River, Guizhou Province, in southwestern China. It is one of the most important power sources in the West-East Power Transmission Project. The Goupitan hydropower project includes 5 large-size units with a total capacity of 3000 MW. Each unit has individual multiple vibration zones. Unit #2 is a unit with typical vibration zones, as indicated in Fig. 5. The Goupitan hydropower project has a multi-year regulating storage reservoir so that the fore-bay water level is relatively fixed during daily operations. When responding to the peak load demand, the Goupitan's tailrace water level varies greatly, with an amplitude of 7–8 m. Developing a real-time operation model that incorporates the head effect on unit performance curves and multiple vibration zones is essential to optimizing the hourly schedules of the Goupitan hydropower project. Table I gives the main characteristics of the Goupitan hydropower project.

Here, we consider one day with 24 hourly periods in the flood season when the Goupitan hydropower project responds to the peak load demand. This is done to test the model in a situation where large variations in net head occur during a single day. Since no spillage is expected, we set $s_t = 0$ to accelerate convergence. Forecasted inflows and forecasted load demands are shown in Table II. The initial fore-bay water level is 602.94 m. The minimum online and offline times are set to be 4 hours and the startup time limit is assumed to be 2 over the whole planning horizon. We set a time limit of 1200 seconds as the stopping criterion to make a quick decision in practice, which later proves to be perfectly adequate.

TABLE II
FORECASTED LOAD DEMANDS AND INFLOWS

Period	1	2	3	4	5	6	7	8
Load demand (MW)	215	215	215	215	215	215	484	702
Inflows (m^3/s)	1,735	765	1,389	1,595	1,184	1,180	989	839
Period	9	10	11	12	13	14	15	16
Load demand (MW)	1,006	1,973	2,488	2,475	2,465	1,811	1,806	1,844
Inflows (m^3/s)	1,005	645	1,043	1,937	1,723	1,497	1,943	1,312
Period	17	18	19	20	21	22	23	24
Load demand (MW)	1,808	2,500	2,493	2,489	2,487	2,486	2,488	2,016
Inflows (m^3/s)	1,946	1,161	1,948	2,153	1,320	1,736	2,153	1,695

TABLE III
NET HEAD DISCRETIZATION

No.	1	2	3	4
Net head interval (m)	[160.00, 162.00]	[162.00, 165.00]	[165.00, 167.50]	[167.50, 173.00]
Pre-specified unit performance curve of # m	162.00	164.00	166.00	170.00

The net head is first estimated by using plant performance curves, ranging from 160.83 m to 172.05 m. In this way, fewer variables are used so that the computation time can be reduced.

Table III shows the net head discretization and unit performance curves simplification.

Table IV provides the upper and lower bounds of each vibration zone in each net head interval.

Table V shows the optimization results. The final gap is 0.53%, meaning that the objective solution is close to the global optimal solution. We believe that the result is acceptable because of the large number of variables in the case study. Furthermore, it is worth noting in particular that the longer run may only tighten the bounds but not improve the objective function, as concluded by [34].

Fig. 6 plots the net head of unit #2 over the entire planning horizon. The net head varies from 159.99 m to 172.91 m, similar to the estimated values using the plant performance curves. The head effect directly impacts power output and multiple vibration zones that cannot be overlooked. The mathematical formulation presented considers head variation over the operation periods.

The units' on/off states in each period are shown in Table VI. Since constraints (15) and (16) are imposed, no unreasonable "flip-flop" occurs in this scenario.

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TABLE IV
UPPER AND LOWER BOUNDS OF EACH VIBRATION ZONE IN EACH NET HEAD INTERVAL

# of Net head interval			1	2	3	4
Unit #1	Vibration zone 1 (MW)	Lower Bound	215	$3.09\times h_{1,t}$-285		232
		Upper Bound	295	$5.45\times h_{1,t}$-588		325
	Vibration zone 2 (MW)	Lower Bound	461	$12.18\times h_{1,t}$-1512		528
		Upper Bound	480	$10.91\times h_{1,t}$-1287		540
Unit #2	Vibration zone 1 (MW)	Lower Bound	$21.5\times h_{2,t}$-3268	$-39.10\times h_{2,t}$+6547		0
		Upper Bound	$19.5\times h_{2,t}$-2864	$-38.91\times h_{2,t}$+6598		81
	Vibration zone 2 (MW)	Lower Bound	-	$-42.00\times h_{2,t}$+7265		230
		Upper Bound	-	$-26.00\times h_{2,t}$+4692		337
	Vibration zone 3 (MW)	Lower Bound	-	542		
		Upper Bound	-	560		
Unit #3	Vibration zone 1 (MW)	Lower Bound	0	0		0
		Upper Bound	32	$-1.45\times h_{3,t}$+267		24
	Vibration zone 2 (MW)	Lower Bound	210	$3.45\times h_{3,t}$-349		229
		Upper Bound	300	$9.09\times h_{3,t}$-1172		350
	Vibration zone 3 (MW)	Lower Bound	-	542		
		Upper Bound	-	560		
Unit #4	Vibration zone	No vibration zone exists				
Unit #5	Vibration zone	No vibration zone exists				

TABLE V
OPTIMIZATION RESULTS FOR THE GOUPITAN HYDROPOWER PROJECT EXAMPLE

# of variables	# of binary variables	# of constraints	Generator memory used (K)
6,312	1,104	10,080	1,805
Iteration	Objective value ($10^7 m^3$)	Objective bound ($10^7 m^3$)	Final gap (%)
5,390,816	8.74	8.69	0.53

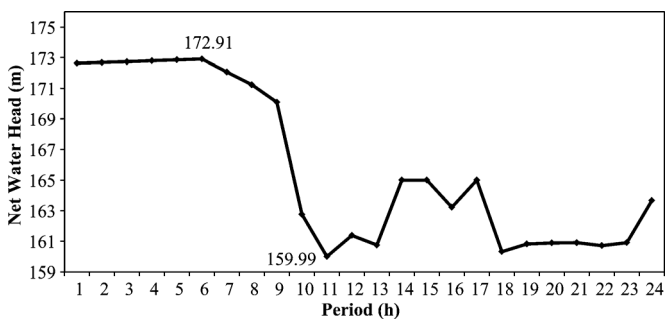


Fig. 6. Net head of unit #2 over the entire planning horizon.

Fig. 7 shows the 24-hour power output schedules of all units belonging to the Goupitan hydropower project.

Unit #1, Unit #2 and Unit #3 are employed to verify that the proposed method ensures that the actual power output avoids all the vibration zones. Based on actual unit performance curves, actual power output is recomputed using the solution of the MILP model. For comparison, another formulation without considering the head effect, using only one unit performance curve of 166 m and regular shaped vibration zones in net head interval of [165.00 m, 167.50 m], is presented here. Actual power outputs from the two formulations are shown in Fig. 8. Actual power output based on the proposed formulation avoids all the vibration zones during the whole planning horizon while it falls into a vibration zone in periods 12, 14, and 24 for Unit #1, 14 and 16 for Unit #2 when the compared formulation is used. This approves that the head effect cannot be overlooked, especially for a unit with multiple vibration zones.

In order to obtain an appropriate stopping criterion, a number of numerical experiments were conducted. The experiments considered 7 situations involving different computation time requirements, which were set to 300, 600, 900, 1200, 1500, 1800 seconds, and no time limit, respectively. Each experiment ran LINGO for 10 times with those time requirements separately. The results are shown in Fig. 9, suggesting that the objective value and final gaps of results differ among experiments,

TABLE VI
UNITS' STATES IN EACH PERIOD

Period \ Unit	#1	#2	#3	#4	#5	Total
1	1	0	0	0	0	1
2	1	0	0	0	0	1
3	1	0	0	0	0	1
4	1	0	0	0	0	1
5	0	0	0	0	1	1
6	0	0	0	0	1	1
7	0	0	0	0	1	1
8	0	0	0	1	1	2
9	1	0	0	1	1	3
10	1	1	1	1	1	5
11	1	1	1	1	1	5
12	1	1	1	1	1	5
13	1	1	1	1	1	5
14	1	1	1	1	1	5
15	1	1	1	1	1	5
16	1	1	1	1	1	5
17	1	1	1	1	1	5
18	1	1	1	1	1	5
19	1	1	1	1	1	5
20	1	1	1	1	1	5
21	1	1	1	1	1	5
22	1	1	1	1	1	5
23	1	1	1	1	1	5
24	1	1	1	1	1	5
Total	20	15	15	17	20	

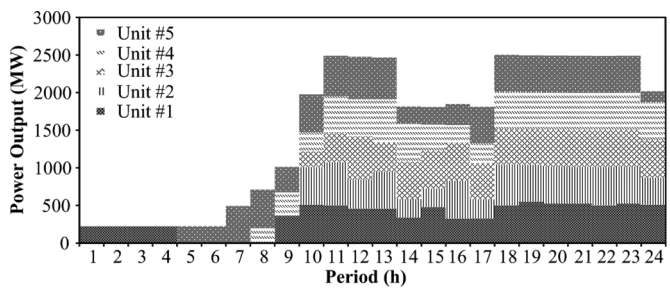


Fig. 7. Power outputs from the Goupitan hydropower project.

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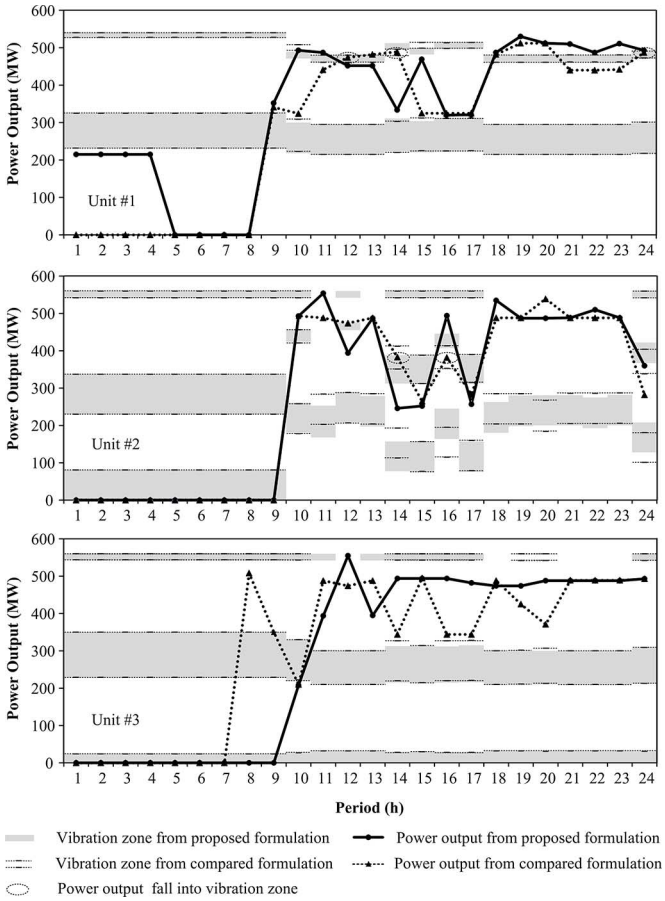


Fig. 8. Actual power outputs from the two compared formulations.

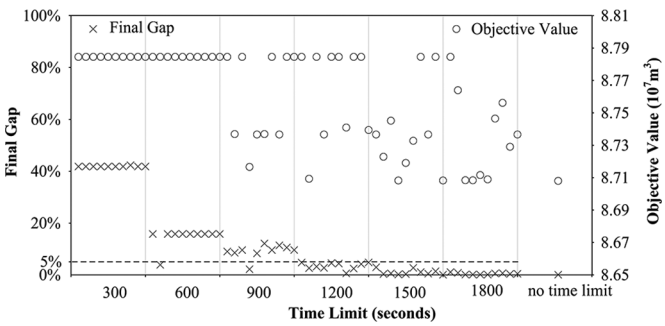


Fig. 9. Results from model runs with different time limits.

even under the same time limit. Furthermore, the objective value and final gap decrease with increased computation time, meaning that the results are approaching to global optimum. As Fig. 9 shows, the final gaps are almost less than 5% with the computation time of 1200 seconds. Thus, 1200 seconds, which was considered as the stopping criterion in this paper, can suitably balance the trade-off between relatively high computational accuracy and low computational time. The case study is further tested with different numbers of safe operating zone. The results are shown in Fig. 10, suggesting that with fewer binary variables, a lower computation time is required to get a better acceptable solution.

Two other scenarios with different load and inflows were also tested. Table VII shows the net head discretization and unit performance curves simplification. Comparison of power outputs

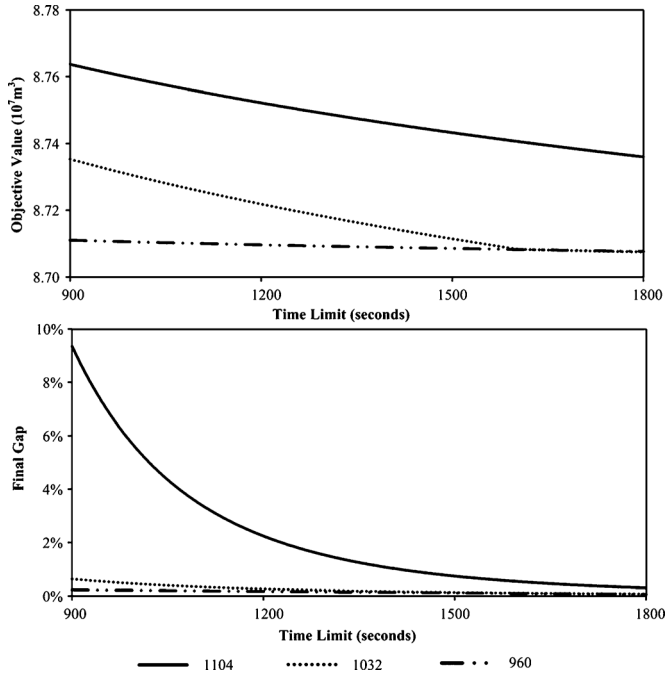


Fig. 10. Trend lines with different binary variables.

are shown in Figs. 11 and 12. In scenario 2, actual power output based on the compared formulation falls into a vibration zone in periods 8, 13, 16, 19, 20, 22, and 24 for Unit #2. In scenario 3, it falls into a vibration zone in periods 16 for Unit #1, 4 and 5 for Unit #2. The two scenarios further tested the validity of the proposed MILP model.

For a single plant and a cascade hydropower system with short term optimal operation, the time steps of 1 hour and 15 minutes are commonly used. Smaller time steps can result in a higher precision. We have tried the proposed MILP formulation with the resolution of 15 minutes, but the computational burden is very high that sometimes the problem cannot be solved within a reasonable time limit using LINGO. Therefore, the resolution of 1 hour is preferred.

While we believe our modeling method is very valuable to the operators that are seriously challenged with the operations of hydropower system with multiple vibration zones, our model is associated with some simplicities and inaccuracies. Thus, it can significantly help the operators improve the current operations without a need for major computational effort. Nevertheless, further improvements to our model are still possible and optimization results can be further improved in the future.

IV. Conclusion

This paper addressed hydro unit commitment with a head-sensitive reservoir and multiple vibration zones using MILP. The MILP formulation mainly deals with constraints of the unit performance curves and multiple irregular shaped vibration zones, which are both closely related to the net head. The unit performance curves are discretized into a set of piecewise curves based on the discretized net head intervals so that the head effect is modeled. The vibration zones are approximated by several simple polygons that depend on the

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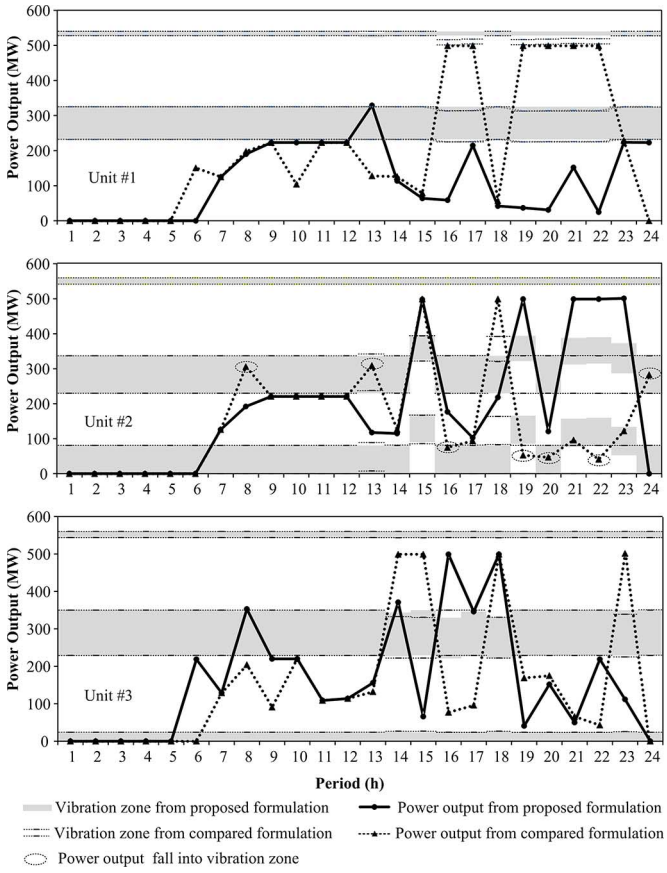


Fig. 11. Actual power outputs from the two formulations in scenario 2.

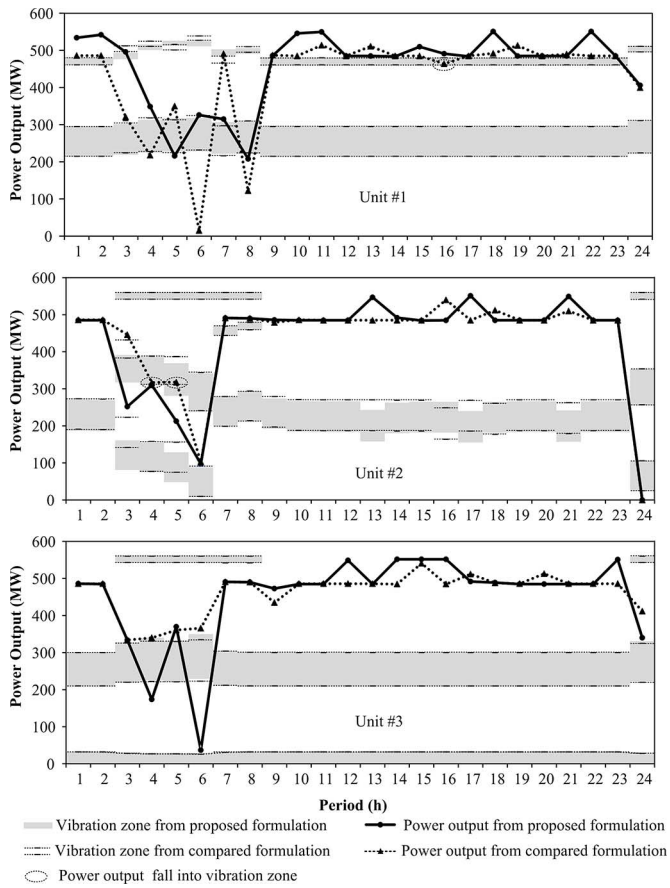


Fig. 12. Actual power outputs from the two formulations in scenario 3.

discretized net head interval so that the head effect is appropriately captured. The Goupitan hydropower plant, one of the most important power sources in West-East Power Transmission Project in China, was used to test the performance and utility of the proposed model. LINGO solved the MILP model and obtained an acceptable optimum in an acceptable computational time. The simulation results demonstrate that unit performance curves linearization and multiple vibration zones linearization used in this paper are suitable for HUC and ensure economic benefits while maintaining the security of hydropower plant. In the future, additional large-scale hydropower projects will be implemented in China; thus, the potential effect of multiple vibration zones on HUC will become more significant. We believe that the proposed model can set a basis for optimal operation and management of hydropower in China.

APPENDIX

A generalized MILP formulation for head-sensitive vibration zones is described in this section. Additional parameters are introduced:

$\overline{NH}_{i,c}$	Right endpoint of net head on unit i for interval c [m].
$\overline{FU}_{i,c,k}$	Right endpoint of safe operating zone k on unit i for interval c [MW].
$\sigma_{i,t,c}$	Binary variable that is equal to 1 if $h_{i,t}$ is in interval c and 0 otherwise.

TABLE VII
NET HEAD DISCRETIZATION FOR SCENARIOS 2 AND 3

Scenario	No.	1	2	3
2	Net head interval (m)	[166.00, 167.50]	[167.50, 169.00]	[169.00, 172.00]
	Pre-specified unit performance curve of # m	166.00	168.00	170.00
3	Net head interval (m)	[160.00, 162.00]	[162.00, 165.00]	[165.00, 167.50]
	Pre-specified unit performance curve of # m	162.00	164.00	166.00

$val_{i,t,c}$	Net head on unit i in period t for interval c [m].
$\overline{f}_{i,t,k}, \underline{f}_{i,t,k}$	Upper and lower bounds of safe operating zone k of unit i in period t [MW].

Using piecewise linearization, the upper and lower bounds are formulated as shown in (45)–(50) at the top of the next page.

The avoidance of vibration zones is modeled through a linear formulation. A major advantage of this formulation is that constraints are independent of the form of bounds. See (51)–(53) at the top of the next page.

In order to describe the piecewise bounds using MILP, additional integer variables are adopted. However, these integer variables impose extensive computational burden due to the MILP model's sensitivity to the number of integer variables.

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$$\sum_{c\in C}\sigma_{i,t,c}=1\ \ \forall i,\forall t \tag{45}$$

$$\sigma_{i,t,c}\in\{0,1\}\ \forall i,\forall t,\forall c \tag{46}$$

$$\sigma_{i,t,c}\times\overline{NH}_{i,c-1}\le val_{i,t,c}\le\sigma_{i,t,c}\times\overline{NH}_{i,c}\ \forall i,\forall t,\forall c \tag{47}$$

$$\sum_{c\in C}val_{i,t,c}=h_{i,t}\ \ \forall i,\forall t \tag{48}$$

$$\overline{f}_{i,t,k}=\sum_{c\in C}\left(\sigma_{i,t,c}\times\overline{NH}_{i,c-1}+\frac{\overline{FU}_{i,c,k}-\overline{FU}_{i,c-1,k}}{\overline{NH}_{i,c}-\overline{NH}_{i,c-1}}\times\left(val_{i,t,c}-\sigma_{i,t,c}\times\overline{NH}_{i,c-1}\right)\right)\ \forall i,\forall t,\forall k \tag{49}$$

$$\underline{f}_{i,t,k}=\sum_{c\in C}\left(\sigma_{i,t,c}\times\overline{NH}_{i,c-1}+\frac{\overline{FD}_{i,c,k}-\overline{FD}_{i,c-1,k}}{\overline{NH}_{i,c}-\overline{NH}_{i,c-1}}\times\left(val_{i,t,c}-\sigma_{i,t,c}\times\overline{NH}_{i,c-1}\right)\right)\ \forall i,\forall t,\forall k \tag{50}$$

$$(\underline{f}_{i,t,k}+\delta_i)-r_{i,t,k}\times M_1\le p_{i,t}\le(\overline{f}_{i,t,k}-\delta_i)+r_{i,t,k}\times M_1\ \forall i,\forall t,\forall k \tag{51}$$

$$\sum_{k\in|K_i|}r_{i,t,k}=|K_i|-g_{i,t}\ \ \forall i,\forall t \tag{52}$$

$$r_{i,t,k}\in\{0,1\}\ \forall i,\forall t,\forall k \tag{53}$$

Nevertheless, the generalized formulation is valuable and practical.

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